

1. Streaming model and compact state

1. Model

- Stream $S = (a_1, a_2, \dots)$ from universe U .
- State update: $M_t = \text{update}(M_{t-1}, a_t)$.
- Answer: $\text{query}(M_t)$.
- Goal: one pass, sublinear memory, fast update.

Pitfall. The sketch is not the data. It supports only the intended queries.

2. Cash-register vs turnstile

- Cash-register: $A[j] \leftarrow A[j] + I, I \geq 0$.
- Turnstile: $A[j] \leftarrow A[j] + U, U$ may be negative.
- Use the right model: counts of arrivals vs active-flow updates.

2. Randomized summaries

1. Reservoir sampling

- Sample k items uniformly from unknown-length stream.
- First k items in reservoir.
- For item $i > k$: keep with prob. $\frac{k}{i}$, evict random reservoir item.
- Guarantee: each item ends with prob. $\frac{k}{n}$.

2. Morris counter

- Maintain counter X , estimate $\hat{n} = 2^X - 1$.
- Increment X with prob. $\frac{1}{2^X}$.
- Space: $O(\log \log n)$ bits.

- Unbiased, but high variance; average multiple counters.

Pitfall. Unbiased does not mean low variance.

3. Hashing and sketches

1. Universal hashing

- Family h with $\Pr[h(x) = h(y)] \leq \frac{1}{m}$ for $x \neq y$.
- Use for randomization, load balancing, sketching.

2. Bloom filter

- Bit array size m , k hash functions.
- Insert: set all k positions.
- Query: all bits 1 $\Rightarrow$ possibly present.
- False-positive rate: $(1 - e^{-k\frac{n}{m}})^k$.
- Optimal $k^* = (\frac{m}{n}) \ln 2$.
- One-sided error: no false negatives.

3. Count-Min sketch

- Update all d rows: $C[i][h_{i(x)}] \leftarrow C[i][h_{i(x)}] + 1$.
- Query: $\hat{f}(x) = \min_i C[i][h_{i(x)}]$.
- Overestimates only.
- Guarantee: $\Pr(\hat{f}(x) - f(x) > \varepsilon N) \leq \delta$.
- Width/depth: $w = \lceil \frac{e}{\varepsilon} \rceil, d = \lceil \ln(\frac{1}{\delta}) \rceil$.

Pitfall. CMS error is additive and depends on stream mass N ; Bloom filters answer membership, not counts.

4. Inference and estimation

1. Likelihood / MLE

- Likelihood: $\mathcal{L}(\theta; x) = P(X = x \mid \theta)$ or density value.
- MLE: $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \mathcal{L}(\theta; x)$.
- Log-likelihood: $\ell(\theta) = \log \mathcal{L}(\theta; x)$.
- Typical steps: product $\rightarrow$ log $\rightarrow$ derivative $\rightarrow$ solve.

2. Common MLEs

- Bernoulli(p): $\hat{p} = \bar{X}$.
- Gaussian μ known σ^2 : $\hat{\mu} = \bar{X}$.
- Gaussian σ^2 known μ : $\hat{\sigma}^2 = (\frac{1}{n}) \sum_i (X_i - \mu)^2$.
- Gaussian both unknown: $\hat{\mu} = \bar{X}, \hat{\sigma}^2 = (\frac{1}{n}) \sum_i (X_i - \bar{X})^2$.

3. Bayesian estimation

- Posterior: $\pi(\theta \mid x) = \frac{\mathcal{L}(\theta; x)\pi(\theta)}{\int \mathcal{L}(\theta'; x)\pi(\theta') d\theta'}$.
- MAP: $\arg \max_{\theta} \pi(\theta \mid x)$.
- Posterior mean: $E[\theta \mid x]$.
- Conjugacy:
 - Beta-Bernoulli: $\text{Beta}(\alpha + k, \beta + n - k)$.
 - Gaussian-Gaussian (known variance): $\sigma_n^2 = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}\right)^{-1}, \mu_n = \sigma_n^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{\sum X_i}{\sigma^2}\right)$.

Pitfall. MLE uses data only; MAP adds the prior; posterior mean minimizes squared error.

5. Estimator quality and decision

1. Risk / error notions

- Loss $L(\theta, a)$ measures cost of action/estimate a when truth is θ .
- Risk = expected loss over data / randomness.
- Bias = $E[\hat{\theta}] - \theta$.
- MSE = bias² + variance.
- Consistency: $\hat{\theta}$ converges to θ as $n \rightarrow \infty$.

2. Hypothesis testing / decision

- H_0 vs H_1 .
- Type I error: reject true H_0 .
- Type II error: fail to reject false H_0 .
- p-value: $P(\text{statistic at least as extreme as observed} \mid H_0)$.
- Likelihood-ratio test compares max likelihoods under H_0 and H_1 .
- Choose actions by minimizing expected loss, not by p-value alone.

Pitfall. A small p-value is evidence against a model, not a guarantee of the real-world claim you want.